

Final Project: Recommendations to Improve Network Security

Estimated time needed: 40 minutes

Learning objectives

After completing this lab, you will be able to:

- Propose security enhancements based on the infrastructure evaluation
- Identify common web application security vulnerabilities based on OWASP guidelines
- Recommend solutions to address basic OWASP security concerns

The final project is divided into two parts:

- First, you will evaluate the existing infrastructure and recommend security enhancements for a scenario.
- Then, you will be presented with a set of questions related to the scenario and your recommendations. The responses to these questions will help you determine if your recommendations were accurate. Additionally, these questions will be part of a graded quiz.

Scenario

Jackson Corporation is experiencing a surge in cyberthreats. The company has hired you to evaluate its existing infrastructure and recommend security enhancements.

After reviewing the existing network and processes, you have identified various concerns. The table summarizes these concerns by component.

Network Component	Findings
1. Firewall	Currently, Jackson Corporation has only one firewall that indiscriminately filters both external traffic from the internet and internal traffic from within the organization. This setup poses a potential risk as it does not allow for targeted security protocols based on the traffic source.
2. Web servers	Jackson Corporation's web server is located internally within the organization's primary network. This setup means that the server, a crucial and sensitive asset, is exposed to the same potential risks as the rest of the internal network. In terms of security, a basic firewall is the only mechanism in place for protecting the web server. This singular firewall is tasked with filtering both incoming external traffic and internal traffic within the network, creating a potential vulnerability.
3. Network monitoring	Jackson Corporation's network infrastructure currently consists of wired and wireless networks. All devices, including employee workstations, servers, and other digital assets, are interconnected within this network, facilitating seamless data exchange and collaboration. There is no comprehensive system to centrally monitor, log, and analyze the security events happening across the entire network. While this interconnected setup allows for operational efficiency, it also opens potential vulnerabilities that internal or external threats can exploit. Currently, there is no comprehensive system in place. This limitation makes threat detection and incident response slower and less efficient, leaving the corporation at increased risk.
4. Breach detection	As it currently stands, Jackson Corporation primarily relies on a basic network monitoring system. This system monitors the traffic and the network's performance but lacks an advanced threat detection mechanism. The current network monitoring setup involves regular checks on the network's health, performance statistics, and traffic volume. However, it does not provide detailed analysis or real-time alerts about potential security threats, anomalies, or suspicious activities within the network. While this setup allows for maintaining basic network performance and identifying bandwidth, latency, or server downtime issues, it falls short in proactively identifying and mitigating potential security threats. As a result, the company might not be able to detect a cyberthreat until after a breach has occurred. This reactive approach to network security puts the corporation's sensitive data and digital assets at significant risk.
5. Remote work	As Jackson Corporation expands globally, its workforce becomes increasingly remote and mobile. Employees frequently travel or work from home, needing to access the corporation's internal resources from different parts of the world. Users are currently using shared resources using web-based tools. However, accessing these resources over public or unsecured networks poses significant security risks. The data exchanged could be intercepted and compromised by malicious entities. In this context, the absence of a secure remote access solution could potentially limit the productivity and efficiency of the corporation's mobile workforce while also exposing the corporation's internal resources to unnecessary security risks.
6. Software development	Jackson Corporation currently follows a traditional waterfall approach to software development. The coding process begins with clear-cut requirements that the development team translates into functional code. Developers code in isolation, focusing primarily on the software's functionality, with little consideration for potential security vulnerabilities. Code reviews are not a regular practice and are typically only conducted on an ad hoc basis when a problem arises. The final product is then passed on to a separate team for testing before it's ready for deployment. The current process creates gaps in understanding and practicing secure coding principles. Without focusing on security from the outset and regular code reviews, vulnerabilities could be overlooked and make their way into the deployed software. This could potentially provide an avenue for cyberthreats to compromise the application and the data it handles.
7. Web application security (OWASP Concerns)	During the security audit, several common web application vulnerabilities were discovered in Jackson Corporation's customer portal and internal web applications. These issues align with common OWASP security risks.

Issue A: Weak password policy - The system allows users to create simple passwords like "password123" or "12345678". There are no requirements for password complexity (such as requiring uppercase letters, numbers, or special characters), and there is no minimum length requirement beyond 6 characters.

Issue B: Unencrypted sensitive data - Customer credit card information and social security numbers are stored in the database in plain text format (not encrypted). If an attacker gains access to the database, they can easily read all this sensitive information.

Issue C: No input validation - The search box and contact forms on the website accept any type of input without checking or cleaning it. Users can type special characters and code into these fields, which could allow attackers to inject malicious code into the system.

Issue D: Detailed error messages - When something goes wrong on the website, error messages display detailed technical information including database names, file paths, and system configuration details. This information could help attackers understand how the system works and find ways to exploit it.

Exercise

Based on the findings listed in the scenario, you now need to provide your recommended solutions to improve the network security.

Document your recommendations for the following:

1. Firewall
2. Web servers
3. Network Monitoring
4. Breach detection
5. Remote work
6. Software development
7. Web Application Security (OWASP Concerns) - For each issue identified (A, B, C, and D), propose a practical solution

Guidelines for OWASP recommendations

For component #7, address each of the four issues:

Issue A (weak password policy): Recommend specific password requirements and additional security measures.

Issue B (unencrypted sensitive data): Suggest how sensitive data should be protected in the database.

Issue C (no input validation): Explain what should be done to user inputs before processing them.

Issue D (detailed error messages): Describe what type of error messages should be shown to users instead.

Tip: Think about simple, practical solutions. For example, for weak passwords, you might recommend requiring passwords to be at least 12 characters long with a mix of letters, numbers, and symbols.

Document recommendations

You can document your recommendations in the attached document.

To download the document, right-click [here](#). Then, click **Open link in a new tab**.

Author(s)

- [Ramanujam Srinivasan](#)



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